EGAFNet: An Edge Guidance and Scale-Aware Adaptive Fusion Network for Building Extraction From Remote Sensing Images

Mingwang Yang, Like Zhao, Linfeng Ye, Weishi Jia, Huawei Jiang, and Zhen Yang

***Abstract*— Accurately extracting buildings from high-resolution remote sensing images is crucial for urban planning, smart city construction, map updating, and other fields. However, the buildings extracted from remote sensing images still have deficiencies such as fuzzy boundaries and poor integrity due to the occlusion of trees and shadows, the interference of redundant information, and the characteristics of buildings with different shapes and scales. In this article, a novel building extraction model edge guidance adaptive fusion network (EGAFNet) is proposed for remote sensing images, which utilizes EfficientNetV2 as the encoder and multiscale feature enhancement module (MFEM) as the decoder to improve the accuracy and efficiency. To address the issue of missing boundary information, a branch for extract boundary features and an edge guidance module (EGM) are constructed to enhance the network's ability to express the boundary. In addition, a scale-aware adaptive fusion module (SAFM) is designed to adaptively aggregate the multiscale features to enhance the network's ability to capture the different scale features. We also adopt a multilayer supervision strategy that predicts the outputs for each layer of the decoding phase and imposes multilevel loss constraints to achieve fast fitting of the model. Experiments are conducted on the WHU building dataset and a dataset of building instances of typical cities in China. The experimental results demonstrate that the EGAFNet achieves the best *F*1-score and intersection over union (IoU) compared with other typical building extraction methods. It suggests that EGAFNet can effectively improve the learning ability of the boundary information and enhance the multiscale building feature representation, thus realizing more accurate building extraction. The code will be available at https://github.com/Mw-yang/EGAFNet.**

***Index Terms*— Building extraction, deep learning, edge guidance, multiscale features, remote sensing images.**

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The authors are with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou 450001, China (e-mail: ymw@stu.haut.edu.cn; like_zhao@haut.edu.cn; Linfengye2021@163.com; gurdian@stu.haut.edu.cn; lhwcad@126.com; 593797596@qq.com).

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I. Introduction

BUILDINGS, being one of the essential infrastructures in urban areas, serve as the primary locations for human activities. They play not only a vital role in human life and development but also in the sectors of the economy, military, and emergency disaster response [1], [2]. In this case, the accurate acquisition of building information has important application prospects for smart city construction, urban planning, and other fields. Along with the rapid development of Earth observation technology, high-resolution remote sensing images have become an important source of building information owing to the clear texture details, rich image layers, and so on. The grayscale features, geometric structure, and texture information presented in remote sensing images are important for building identification and extraction [3], [4]. Therefore, how to extract building information efficiently and accurately from remote sensing images has become a hotspot and a challenge for scholars [5], [6].

Although building information can be obtained from remote sensing images through manual visual interpretation, this method is inefficient and time-consuming. Therefore, it becomes crucial to develop efficient and accurate building extraction methods. In the past decades, many researchers at home and abroad have conducted in-depth studies on building extraction methods, and according to the different ways of feature extraction, the methods for building extraction mainly include two categories: traditional manually designed feature extraction methods and deep learning methods.

The traditional manually designed feature extraction methods mainly extract the features (such as texture, shape, and spectrum) by manually formulating the feature rules of the building [7] and then identify buildings using machine learning algorithms (such as support vector machine and random forest) for feature classification [8] and combining them with the traditional image processing methods (such as filtering, grayscale transformation, and histogram equalization). For example, Wu et al. [7] combined spectral and shape features to address the issue of spectral feature diversity, constructed corresponding templates based on the shape regularity of buildings, segmented the image with the OTSU algorithm, and utilized morphological dilation operations and the Canny operator to achieve further optimization. Huang and Zhang [9] designed the morphological building index

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(MBI) and the morphological shadow index (MSI) for the automatic extraction of buildings in high-resolution images. Based on this, Bi et al. [10] proposed a multiscale filtering building index (MFBI) for building extraction, which can overcome the disadvantage of high computational effort in the morphological operations of MBI. Lv et al. [11] used the orientation gradient features in remote sensing images for the preliminary identification of buildings and then filtered the nonbuilding information in the preliminary extraction findings using features such as texture and spectra to achieve better extraction results. Although the above traditional methods can effectively utilize the spectral, texture, shape, and other features of remote sensing images with strong interpretability, their performance fluctuates significantly in diverse situations due to variations in sensors, building types, and ground conditions. Consequently, these methods are only applicable to a small range of image data with similar features and have poor generalization ability when dealing with complex backgrounds, variable lighting conditions, and diverse building types.

With the rapid development of deep learning, neural networks have been widely used in various tasks in the field of computer vision due to their excellent feature capturing ability, such as image classification [12], object detection [13], semantic segmentation [14], and instance segmentation [15]. In recent years, some scholars have also begun to apply them to building extraction tasks and have achieved remarkable results [16]. For example, Luo et al. [17] designed Dense ASPP to enhance the extraction ability of small and irregular buildings by modifying the original five branches from parallel to series and connecting them in a denser form. However, the boundary jaggedness of this method is obvious, which may have some influence on the extraction results. In order to deal with the problem of large changes in building scale, Zhang et al. [18] added original image information of different scales to the shallow encoder and combined the asymmetric convolution and the dilated convolution with different dilation rates to achieve the goal of capturing multiscale contextual features without significantly increasing the number of parameters. Yu et al. [19] designed the feature pyramid attention (FPA), which utilizes convolutional kernels of different sizes to fuse different scale features layer by layer, effectively avoiding feature loss during the sampling process and achieving good performance. In addition, Gong et al. [20] proposed C^3Net, which inputs the multiscale features output from ASPP to the context-aware module, excavates the spatial context information to improve the multiscale feature representation, and further fuses the building boundary information obtained by the Sobel operator based on the residual refinement module, thereby improving the completeness of the building extraction. Qiu et al. [21] proposed MSL-Net, which replaces the second BN layer in each inverted residual block of MobileNetV2 with a GN layer to avoid model degradation caused by dataset distribution deviation, followed by ASPP and deformable convolution to adapt to the diversity of building shapes. This method achieves a good balance between parameters and accuracy. However, it still has certain limitations when facing more complex scenes.

To sum up, the method based on deep learning has a higher degree of automation and better generalization ability than traditional methods and has good performance in terms of accuracy and completeness. Although many previous studies have been conducted, there are still issues such as omission and erroneous extraction due to the inability to fully exploit the building's multiscale features, as well as fuzzy and rough boundaries caused by occlusion and noise interference [22]. To address these issues and enhance the accuracy of building extraction, it is necessary to improve the neural network's ability to extract multiscale building features, thereby enhancing the network's generalization ability, and to optimize the building modeling process by considering the building boundary contour constraints. Aiming to solve the above problems, this article proposes a novel edge guidance adaptive fusion network (EGAFNet).

The contributions of this article are summarized as follows.

1) In the encoding stage of EGAFNet, the EfficientNetV2 network is introduced to extract multilevel hierarchical features. At the same time, a multiscale feature enhancement module (MFEM) is constructed as a decoder module to fully explore the deep semantic information of the image and help the network to extract the target information of the building more accurately in the decoding stage.
2) Aiming at the problem of the lack of boundary information, the edge branch (EB) and edge guidance module (EGM) for learning boundary features are constructed. The acquired boundary information is utilized to guide the image restoration process, compensating for the deficiency in boundary information within the decoder's spatial modeling. The EB and EGM can enhance the network's ability to express building boundaries, thereby optimizing the effect of building boundary extraction.
3) The scale-aware adaptive fusion module (SAFM) is designed to fully excavate the multiscale features of buildings and suppress noise interference while realizing the adaptive aggregation of multiscale features within the image. The SAFM module effectively enhances the network's perception of different scales and strengthens the model's robustness, thereby addressing the issues of missed and incorrect extraction of buildings.
4) To achieve efficient updating of the decoder layer parameters and to supervise the fast fitting of the model to the target building information, the multilayer supervision module (MSM) is introduced, which predicts the output of features at each layer of the decoding stage and imposes multilevel auxiliary loss constraints.

II. Method

EGAFNet mainly consists of four parts: encoder, decoder, SAFM (located between the encoder and decoder), and EB (separate branch). In this section, we first provide an overview of EGAFNet and then introduce the EB, SAFM module, MFEM module (decoder feature extraction module), and MSM module (located in the decoder layers D_4, D_3, and D_2).

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A. Overview

The proposed EGAFNet is an end-to-end network with an encoder–decoder structure, as shown in Fig. 1. Specifically, EGAFNet adopts EfficientNetV2 [23] as the encoder and the four levels of feature maps (E_1–E_4) are extracted layer-by-layer after first downsampling through Stem Conv. Among them, E_1 and E_2 are low-level features with high resolution and rich detailed information, but less semantic information. While E_3 and E_4 belong to high-level features, although they have obvious semantic information, the loss of detail information is serious.

In order to obtain building boundary features with rich details, EGAFNet uses the EB to effectively aggregate low-level features E_1, E_2, and high-level feature E_3, and then, the EGM module guides the boundary information to each layer of the decoder, enhancing the network's ability to express the boundary during the decoding process.

In addition, to enhance multiscale feature representation, the SAFM module is introduced between the encoder and decoder. It utilizes an attention-gate unit to filter irrelevant features and aggregate feature maps of different scales layer by layer to improve the feature capture ability of the model at different scales.

In the decoder part, the MFEM module is designed as the decoder of each layer. It utilizes the convolution operation with different kernel sizes to strengthen the multiscale feature modeling ability in the decoding process. The feature map is restored to the original size by the four-layer decoder and upsampling to obtain the building extraction results. Meanwhile, we designed the MSM module to impose multilevel joint constraints on the decoding layers D_2, D_3, and D_4, which further improved the accuracy of building extraction.

B. EB and EGM

The low-level features retain sufficient building boundary information, which may gradually weaken as it is passed layer by layer to deeper layers [24], [25]. Therefore, to fully utilize the boundary information to improve the blurred and jagged boundary phenomena, this article designs the EB used to learn the boundary features and the EGM to guide boundary features, as shown in Fig. 2(a) and (b).

Specifically, the EB first upsamples E_1 and E_2 and uses convolution to extract features. After splicing in the channel dimension, it further fuses the semantic information contained in E_3. Then, the boundary prediction results are generated using 1×1 convolution and Sigmoid functions. In addition, boundary labels are generated by applying the Canny operator to the building labels, and the edge loss is calculated with the boundary prediction results. The edge loss is back-propagated and the gradient is updated at each epoch, so as to improve the accuracy of the network for boundary pixel recognition.

The EGM takes feature E_{23} from EB as input, downsamples it to the same size of feature Decoder_n, and then splices it along the channel dimension to supplement boundary information for the decoder. Subsequently, the feature is further extracted using depth-separable convolution and connected with the decoder feature residuals to supplement the semantic information of the image. Finally, the feature is upsampled and passed to the next layer of Decoder, Decoder_{n-1}, to further recover the image features.

C. Scale-Aware Adaptive Fusion Module

Buildings usually have different shapes, which makes the shape, texture, and other features presented in the images have more obvious differences [26], [27]. Therefore, how to make full use of the deep semantic information of images to achieve effective aggregation of multiscale features of buildings is a problem worth exploring [28].

Dilated convolution [29] can expand the receptive field of the network, thus solving the above problems to a certain extent. Based on this, the SAFM module is constructed to capture multiscale contextual information and aggregate global information. The structure is shown in Fig. 3(a). In the SAFM module, four parallel dilated convolutions with dilate rates of 1, 2, 4, and 8 are first utilized to capture building features of different scales. Among them, inspired by the attention gate [30], we designed the attention-gate unit to fuse features at different scales, suppressing irrelevant feature expressions. Its structure is shown in Fig. 3(b). The computational process of the attention-gate unit can be expressed as follows:

$$\begin{aligned}\text{Gate}(x_1, x_2) = (x_1 \otimes (\sigma(f^{1\times1}(R(x_1 \oplus x_2)))))\\ \oplus (x_2 \otimes (\sigma(f^{1\times1}(R(x_1 \oplus x_2)))))\end{aligned} \tag{1}$$

where x_1 and x_2 denote the input two features with different dilate rates, $\oplus$ indicates the element-by-element summation operation, $\otimes$ represents the multiplication operation, R refers to the ReLU function, $f^{1\times1}$ denotes the 1×1 convolutional layer, and σ is the sigmoid function.

The features of two adjacent branches in SAFM are input into the attention-gate unit to avoid the interference of non-building pixels on feature extraction and are fused in a cascade in turn. Finally, in order to achieve adaptive fusion of multiscale features, the original features and the fused features are adaptively added by a learnable parameter to automatically adjust the proportion of two features during the network training process. Notably, the weight parameters of the four dilated convolution branches of this module are shared. It can reduce the parameters and learning costs of the model while also minimizing the risk of model overfitting.

Therefore, the output of SAFM can be expressed as

$$F_{\text{out}} = a \cdot \text{Gate}(\text{Gate}(\text{Gate}(F_1, F_2), F_3), F_4) + (1 - a) \cdot F_{\text{in}} \tag{2}$$

where F_{in} is the input feature; F_{out} is the module output result; F_1–F_4 denote the feature maps generated by the dilated convolution with a dilate rate of 1, 2, 4, and 8, respectively; and α represents a learnable parameter that can be used to adjust the adaptive fusion ratio of features F_{34} and F_{in}.

D. Multiscale Feature Enhancement Module

Currently, most networks use continuous 3×3 convolution, BN, and ReLU operations in the decoder to extract features.

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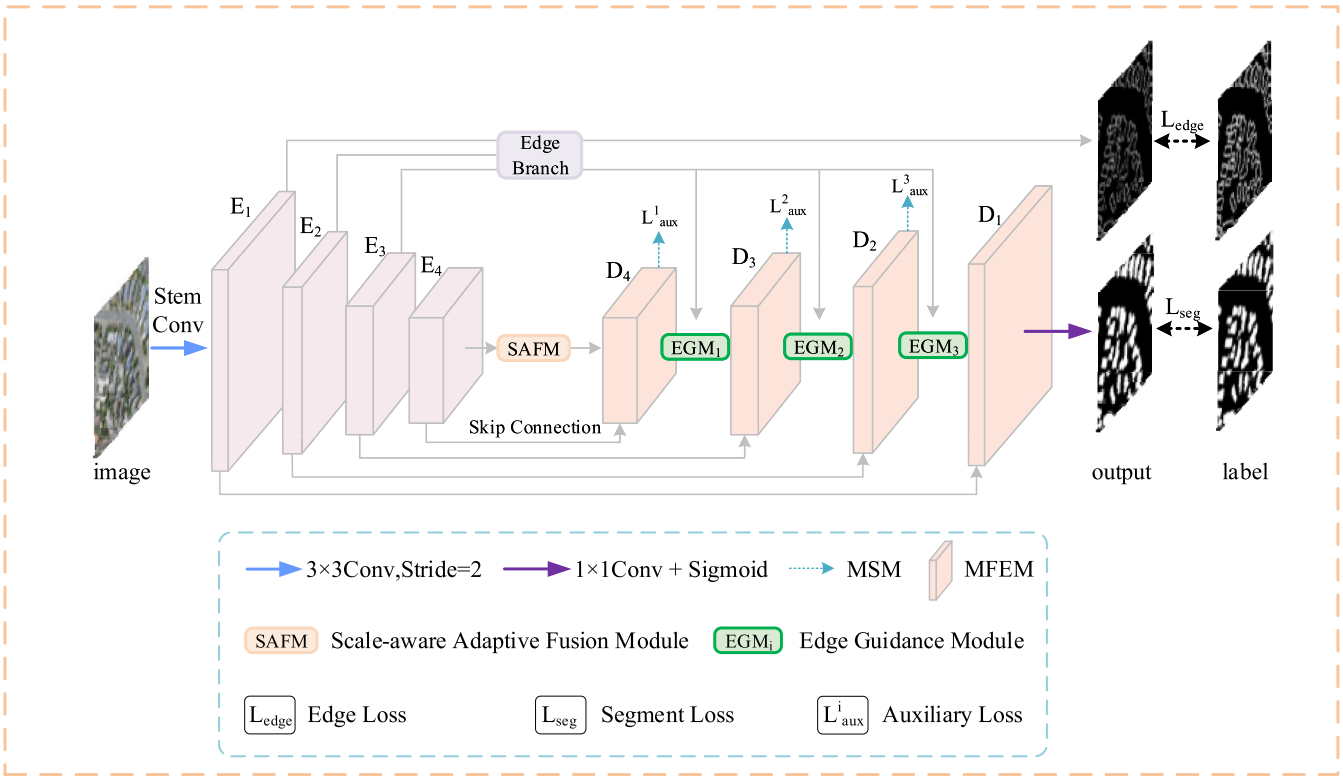


Fig. 1. EGAFNet.

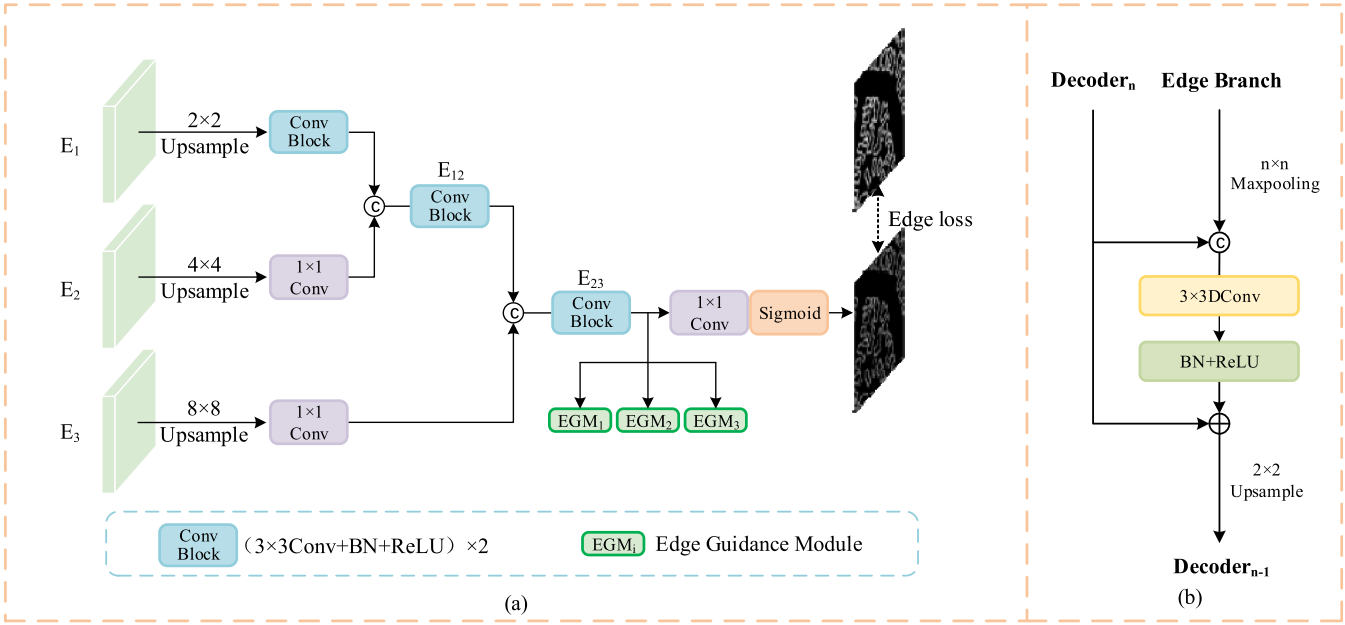


Fig. 2. (a) EB and (b) EGM.

However, simple convolution operations cannot fully excavate the deep semantic information of the image. Therefore, inspired by the structure of inception [31], this article constructs MFEM as the feature decoder module of each layer in the decoding stage to solve this problem, and its structure is shown in Fig. 4.

First, the decoder feature from the previous layer and the encoder feature from the skip connection at the same layer are concatenated as the input of the MFEM module. The module adopts a parallel structure of five branches. The first branch is the residual connection, which is designed to avoid problems such as gradient disappearance during network training. The second branch employs 1×1 convolution for pixel-by-pixel feature extraction to capture the detailed information. The third branch employs 3×3 convolution to capture the information of eight-pixel points around a pixel, fully exploiting the local

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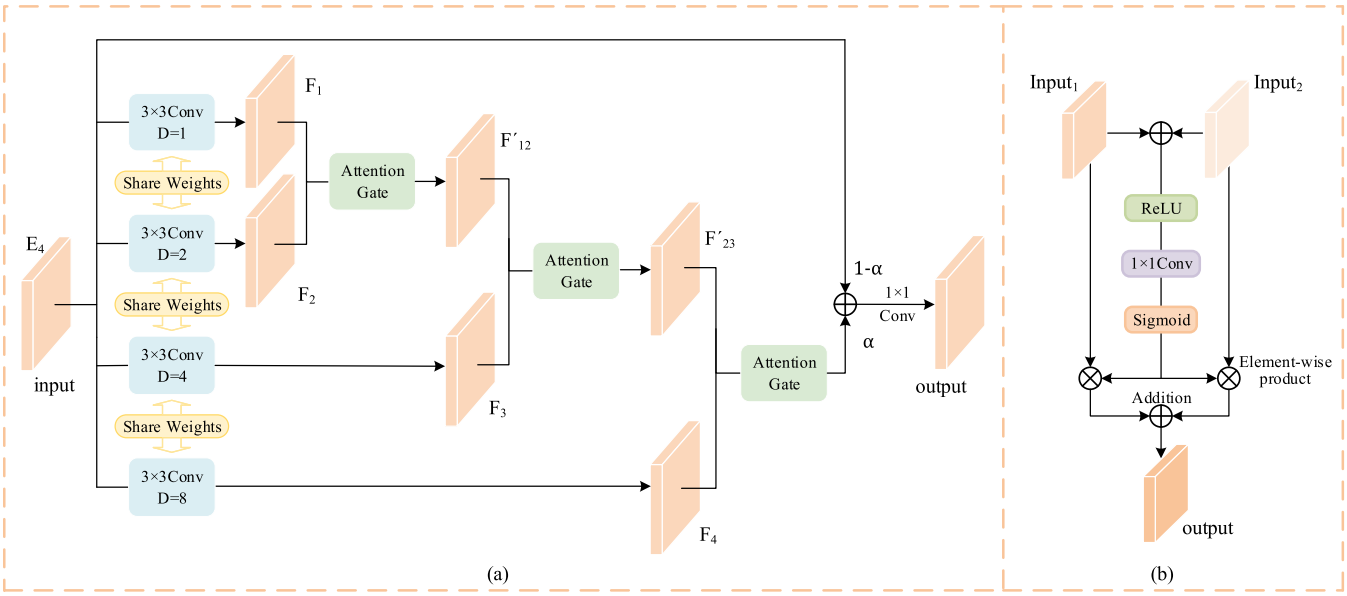


Fig. 3. (a) SAFM and (b) attention-gate unit.

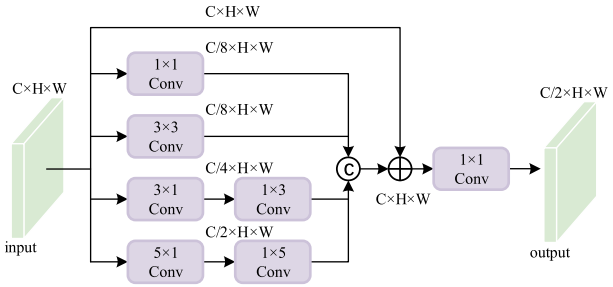


Fig. 4. MFEM.

features of building. The fourth and fifth branches introduce two sets of asymmetric convolutions (3×1 and 1×3, and 5×1 and 1×5), respectively. Asymmetric convolutions of different scales can not only extract features that are easily ignored by square receptive fields but also significantly reduce computational costs. Subsequently, the latter four branches are concatenated and connected with the first branch residual to obtain the output of MFEM.

If x denotes the input of the MFEM module, $f^{n\times n}$ indicates the $n \times n$ convolution operation, C refers to the concatenation operation, and $\oplus$ represents the addition operation, the calculation of MFEM can be represented as follows:

$$F_{\text{out}} = f^{1\times1}(C(f^{1\times1}(x); f^{3\times3}(x) \\ f^{1\times3}(f^{3\times1}(x)); f^{1\times5}(f^{5\times1}(x))) \oplus x). \quad (3)$$

By aggregating features of different scales within five branches, the MFEM module is able to capture wider semantic information in the decoding stage and improve the discriminative ability of multiscale features, thereby helping the network to extract building information more accurately in the decoding stage.

E. MSM and Loss Function

Currently, networks used for building extraction usually generate prediction results only in the last layer, so these methods cannot fully utilize the semantic information from other layers, causing the extraction results to be overly dependent on the last layer features [32]. To address this problem, this article proposes the MSM module, which independently supervises the different feature layers (D_4, D_3, and D_2) in the decoding stage and imposes multilayer constraints, thereby allowing the network to fit the information of the target building faster, as shown in Fig. 5. By performing convolution and upsampling operations on each input feature, the prediction results are generated and the auxiliary loss function L^i_{aux} is calculated with the corresponding truth label.

In order to enhance the supervision ability of the network, three auxiliary losses are introduced into the loss function of the network in a hierarchical form to participate in the backpropagation of the network training and achieve multilevel joint constraints. By independently supervising the middle layer features, the network avoids over-reliance on the last layer's features and effectively leverages the semantic information from the middle layers, thereby enhancing the accuracy of building extraction.

We use the binary cross-entropy loss function (BCE Loss) to calculate the loss. The computation of BCE Loss is shown in the following equation:

$$\text{BCELoss}(p, y) = -\sum_{i}^{N} (y_i \log(p_i) + (1 - y_i) \log(1 - p_i)) \quad (4)$$

where N is the number of pixels and y_i and p_i denote the label and prediction result of the ith pixel, respectively.

In addition, the loss function in this article covers three parts: the building segmentation loss, the loss of EB, and the auxiliary loss. To regulate the impact of each part of the loss

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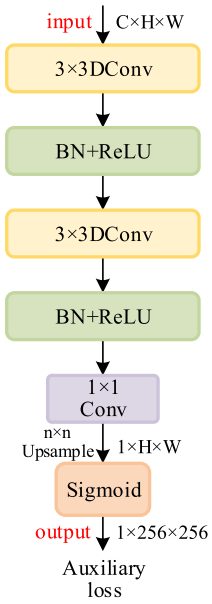


Fig. 5. MSM.

more finely on the overall performance, weighting coefficients are set for each part of the loss function so that the overall loss function can be expressed as

$$L_{\text{total}} = \sum_{i=1}^{3} \lambda_i L_{\text{aux}}^{i} + \lambda_{\text{seg}} L_{\text{seg}} + \lambda_{\text{edge}} L_{\text{edge}} \tag{5}$$

where L_{aux}^{i} refers to the auxiliary loss; L_{seg} denotes the loss of building segmentation; L_{edge} indicates the loss of EB; and λ_i, λ_{seg}, and λ_{edge} represent the respective weight value. In this article, by comparing the experimental results of different weight combinations, λ_1, λ_2, λ_3, λ_{seg}, and λ_{edge} are finally set to $\{0.1, 0.1, 0.1, 0.4, 0.3\}$.

III. Experiment and Analysis

A. Dataset Description

To verify the validity of the proposed model, the WHU building dataset [33] and a dataset of building instances of typical cities in China [34] are selected for experimental analysis.

1) WHU Building Dataset: It has become a very popular dataset in the field of building extraction due to wide coverage, high spatial resolution, and large data volume. It contains two subsets, the aerial imagery dataset and the satellite dataset. Among them, the aerial imagery dataset is more commonly used by researchers. This article also selects this subset for experiments. It covers an area of about 450 km^2 of New Zealand and contains about 220 000 independent buildings. It contains 8189 images of 512 × 512. These are divided into three parts: 4736 for training, 1036 for validation, and 2416 for testing. It is available from http://gpcv.whu.edu.cn/data/building_dataset.html.

2) Dataset of Building Instances of Typical Cities in China: This dataset, constructed by a research team from China University of Geosciences, is the first publicly available dataset composed entirely of remote sensing imagery of cities in China. The dataset contains satellite images of four cities, Beijing, Shanghai, Shenzhen, and Wuhan, with a resolution of 0.29 m, covering a total area of about 120 km^2. It consists of 7260 images of 500 × 500 pixels in size, of which 5985 are for the training set and 1275 are for the test set. Since this dataset is not pre-divided into a validation set, we further split the training set into a training set and validation set with a ratio of 8:2. Consequently, the dataset used in our experiments consists of 4788 images for training, 1197 for validation, and 1275 for testing. The dataset is acquired at https://doi.org/10.11922/sciencedb.00620.

Unlike the WHU building dataset, the images in this dataset are all nonorthophotos, and the dataset covers the main urban areas of China's prosperous cities. Therefore, the deviations of the tilt angle of the buildings and the interference from surrounding redundant features make this dataset more challenging.

B. Experimental Details

All experiments in this article are trained and tested on NVIDIA GeForce RTX 3080 10 GB. EGAFNet and other comparison methods are constructed using the Pytorch framework. Due to limitations such as computational resources, we used random hue saturation values, random scaling, random horizontal–vertical flipping, and random rotation by 90° for data enhancement of the original dataset in the data preprocessing stage, followed by cropping the image to 256 × 256 size as input to the network, with the batch size parameter set to 4. During training, the Adam algorithm [35] is utilized to optimize the network, where the initial learning rate is set to 0.0005 and the learning rate is adjusted using the cosine annealing algorithm [36] with a period of ten epochs, and EGAFNet is trained for 100 epochs on both the WHU building dataset and a dataset of building instances of typical cities in China. The hyperparameters used in the comparison methods in this article are consistent with those described in the original articles.

C. Evaluation Metrics

In this article, we select five evaluation indicators: overall accuracy (OA), precision, recall, $F1$-score ($F1$), and intersection over union (IoU) to evaluate the performance of EGAFNet and other comparison methods. Their formulas are shown in the following equations:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{6}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{10}$$

where TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative, respectively. A larger

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value of these evaluation metrics represents better model performance.

D. Comparison of State-of-the-Art Methods

In this section, we conduct a series of comparative experiments to validate the effectiveness of EGAFNet. Specifically, we compare EGAFNet against nine methods. The selected methods include UANet [37], DSAT-Net [38], MSGCNet [39], EfficientUNet+ [40], C^3Net [20], MMBNet [41], CFENet [42], Res2-Unet [43], and MSL-Net [21]. In our experiments, the best results in each metric are denoted in bold and the second-best results are underlined.

1) Comparison With Other Methods on the WHU Building Dataset: The experimental results of different methods on the WHU buildings dataset are shown in Table I. As seen in Table I, our EGAFNet achieves excellent performance, with four evaluation indicators reaching the best values. Compared to UANet, which achieved suboptimal results in three indicators, the $F1$ and IoU of EGAFNet are 0.44% and 0.77% higher, respectively. Therefore, it can be concluded that the overall performance of EGAFNet is superior to other mainstream methods.

Fig. 6 shows an example visualization of the extraction results of our proposed EGAFNet with other comparison methods on the WHU building dataset. In the visualization results, green represents correctly predicted building pixels, red represents pixels that are actually background but misjudged by the network as buildings, and blue represents pixels that are buildings but missed by the network as background. From the first row, it can be seen that EGAFNet not only does not have a wide range of omission phenomenon of MSGCNet, MMBNet, and Res2-Unet but also reduces the pixel misclassification phenomenon near the boundary of the building dramatically compared to the other methods. This effectively proves that the EB and the EGM module can address boundary blurring and other issues very well. From the results in the second and third rows, it can be seen that for large buildings and complex scenes with varying shapes and scales, our method performs better than other methods in terms of detail information processing because it introduces the SAFM module. For the scene with densely distributed buildings in the fourth row, EGAFNet shows two obvious omissions, and the larger one is that the building is completely covered by trees, which leads to the omission; and the smaller one is a double-sloped-roofed building, where the effect of lighting conditions makes the two sides of the sloped-roofed building show different colors and texture features, and our method can only predict the lighter side of the color and misclassify the darker side as a shadow, thus resulting in omission. It is worth mentioning that these two buildings are omitted in the extraction results of all methods, while other methods still have different degrees of omission except for these two buildings, which further proves that EGAFNet can effectively reduce the occurrence of building omission. The cement vacant area in the fifth row shares similar spectral characteristics with the off-white factory adjacent to it, resulting in DSAT-Net, MMBNet, Res2-Unet, and MSLNet methods to misjudge the cement vacant land as a building. Although the other methods did not misjudge the cement vacant land, they misjudged the surrounding vehicles and other objects with the same spectral features as buildings to varying degrees. In contrast, EGAFNet demonstrates fewer instances of misjudgment, highlighting that the proposed method is more anti-interference and more stable.

The evaluation metrics and visualization results show that EGAFNet can fully explore the semantic information of buildings at different scales, utilize the global contextual features of the image to a larger extent, and use the boundary information to guide the network to carry out the correct building extraction. It enables the network to have more complete boundary information than other methods, especially in complex scenarios such as densely distributed buildings, and the deficiencies of insufficient integrity have also been greatly improved.

2) Comparison With Other Methods on a Dataset of Typical Urban Building Examples in China: To further validate the generalization ability of different methods for buildings with different shapes and scales, experiments are conducted on the dataset of typical urban buildings in China. The experimental results of the different methods are shown in Table II, with the optimal values in each index indicated in bold and the suboptimal values underlined. As shown in Table II, EGAFNet achieved the best results in $F1$ and IoU, and the second-best result in OA. Compared to UANet, which achieved the best OA and second-best $F1$ and IoU, EGAFNet improved $F1$ and IoU by 0.05% and 0.08%, respectively. Although MSGCNet and Res2-Unet achieved the best result in either precision or recall, their other metric was relatively low, resulting in significantly lower $F1$ and IoU than our method, placing them tied for third place (after rounding values are the same). Therefore, it can be concluded from Table II that the generalization ability of EGAFNet is better than that of other compared methods.

Fig. 7 shows the visualization of the building extraction results for different methods and complex scenes for the example dataset of buildings in typical Chinese cities. It can be seen that EGAFNet visually outperforms the other tested methods when dealing with buildings in different complex scene types. As can be seen in rows 1–3 of Fig. 7, the spectrum of the original image is overall dark. Many buildings have the problem of angular shift, and the interference of the surrounding trees and shadows is more serious, which makes this part of the image greatly test the network's ability to extract the occluded buildings. Judging from the extraction results, compared with other methods, the number of red and blue pixels in our method is significantly reduced, the building boundary is smoother and clearer, and there is no issue of misjudging the side elevation of the building as the roof, as seen in the case of DSAT-Net, MMBNet, and so on. It is a good proof that the EGM module and MSM module can guide the network to learn more discriminative features, thus realizing a better building extraction. In the visualization results in rows 4 and 5, in the face of the complex urban scene where buildings are densely distributed and have different shapes and scales, our method has fewer omissions and fewer false extractions than other methods, and the extracted building outlines are clearer, which shows a strong feature capturing

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TABLE I
COMPARISON WITH NINE METHODS ON THE WHU BUILDING DATASET

Method	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
UANet	98.01	**94.50**	93.34	93.91	88.53
DSAT-Net	97.29	91.73	91.81	91.77	84.79
MSGCNet	97.25	91.58	91.73	91.65	84.59
EfficientUNet+	97.02	90.22	91.83	91.02	83.52
C^3Net	97.78	92.71	93.86	93.28	87.41
MMBNet	97.36	91.49	92.57	92.03	85.23
CFENet	96.01	86.28	90.02	88.11	78.75
Res2-Unet	97.00	89.96	92.00	90.96	83.43
MSL-Net	96.62	89.13	90.45	89.78	81.46
EGAFNet	**98.13**	93.82	**94.88**	**94.35**	**89.30**

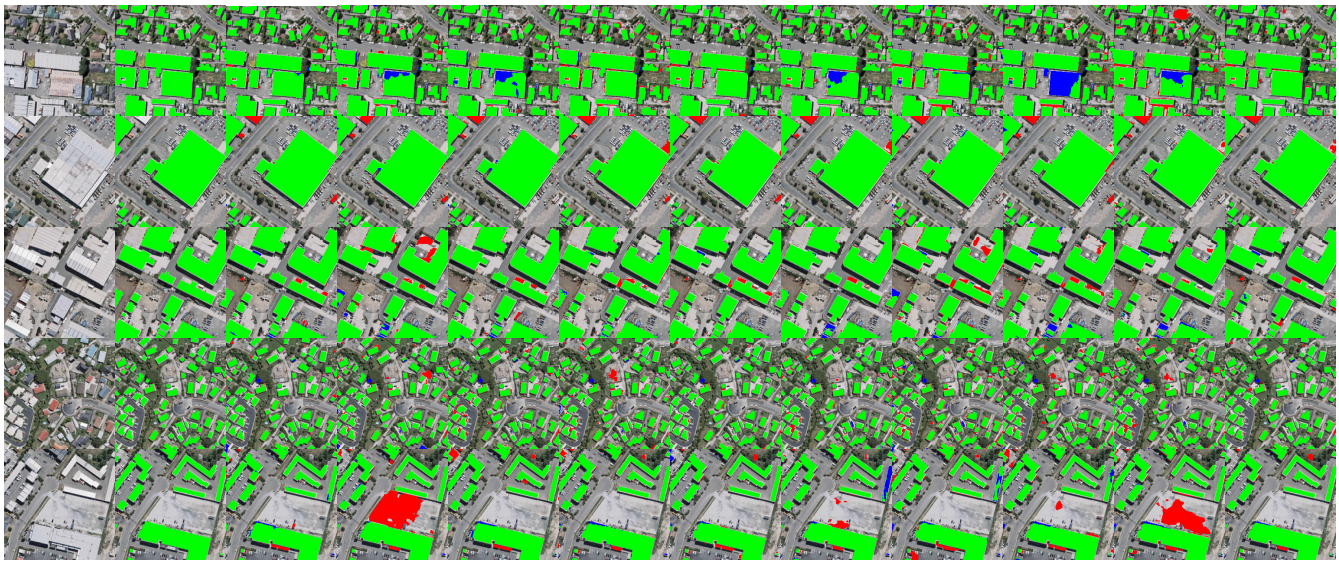


Fig. 6. Example of visualization of building extraction results obtained by different methods on the WHU building dataset. (a) Original image. (b) Label. (c) UANet. (d) DSAT-Net. (e) MSGCNet. (f) EfficientUNet+. (g) C^3Net. (h) MMBNet. (i) CFENet. (j) Res2-Unet. (k) MSL-Net. (l) Ours. (Note: green represents TP, red represents FP, and blue represents FN in the figure).

TABLE II
COMPARISON WITH NINE METHODS ON A DATASET OF TYPICAL URBAN BUILDING EXAMPLES IN CHINA

Method	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
UANet	**93.85**	84.80	87.20	85.98	75.41
DSAT-Net	92.41	82.02	83.08	82.55	70.28
MSGCNet	93.82	**85.17**	86.47	85.81	75.15
EfficientUNet+	92.46	79.29	88.13	83.48	71.64
C^3Net	93.00	81.18	88.02	84.46	73.10
MMBNet	92.84	81.90	85.86	83.83	72.16
CFENet	91.23	76.71	85.35	80.80	67.78
Res2-Unet	93.68	83.30	**88.48**	85.81	75.15
MSL-Net	91.36	78.23	83.16	80.62	67.53
EGAFNet	93.84	84.38	87.75	**86.03**	**75.49**

ability. It indicates that EGAFNet can effectively excavate the multiscale contextual features in the image. At the same time, it can suppress the expression of redundant feature information and further improve the integrity of building extraction.

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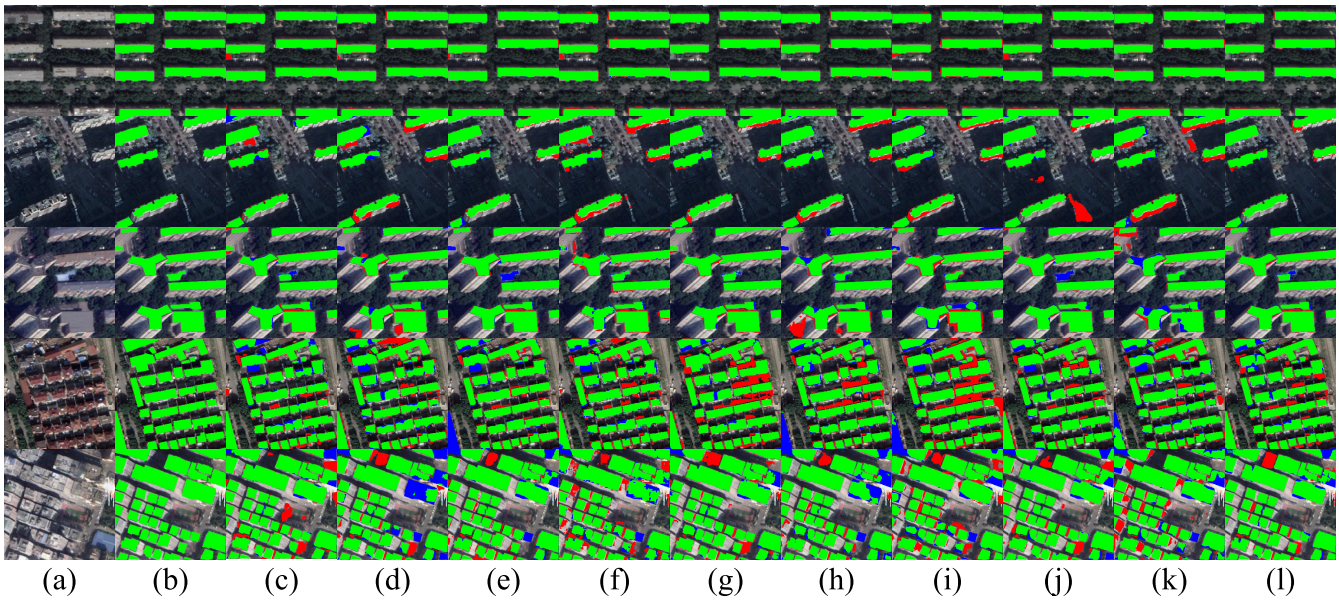


Fig. 7. Example of visualization of building extraction results obtained by different methods on a dataset of typical urban building examples in China. (a) Original image. (b) Label. (c) UANet. (d) DSAT-Net. (e) MSGCNet. (f) EfficientUNet+. (g) C^3Net. (h) MMBNet. (i) CFENet. (j) Res2-Unet. (k) MSL-Net. (l) Ours. (Note: green represents TP, red represents FP, and blue represents FN in the figure).

IV. Discussion

In this section, the effect of each partial module in EGAFNet and the combination of loss function weighting coefficients on the performance of the network are discussed by ablation study in Section IV-A. Section IV-B discusses the model complexity of EGAFNet and other comparative methods.

A. Ablation Study

The ablation studies in this section are divided into three parts: 1) exploring the effect of each module on network performance on the WHU buildings dataset; 2) exploring the effect of each module on network performance on a dataset of typical urban building examples in China; and 3) exploring the effect of the variation of the combination of the loss function weights on the network performance on the two datasets. All experimental conditions are consistent with Section III-B.

1) Performance Analysis of Different Modules on WHU Building Dataset: In this experiment, Backbone denotes a network using only the encoder as EfficientNetV2 and the decoder as the MFEM module. To verify the effectiveness of each module, we integrate each module of SAFM, EB, EGM, and MSM into the Backbone individually and sequentially into the Backbone. The results of the ablation studies on the WHU building dataset are shown in Table III. In addition, because the model must include EB when EGM is included, we change the combination from Backbone ++ EGM to Backbone + EB + EGM.

From Table III, we can see that the individual addition of any module improves Backbone's accuracy. After integrating SAFM into Backbone, the performance can be improved by 0.35% and 0.63% in terms of $F1$ and IoU, respectively. It is because the SAFM module can effectively aggregate the multiscale features of the building, which improves the performance of the network. After adding EB alone, $F1$ and IoU increased by 0.31% and 0.56%, respectively. With the further introduction of EGM, $F1$ significantly increased to 93.91%, indicating a good synergy between EB and EGM, which helps the network better utilize boundary information during spatial modeling. When MSM is added alone, both $F1$ and IoU also elevated, indicating that strong supervision capabilities can improve the accuracy of building extraction. Furthermore, comparing rows 1, 2, 6, 7, and 8, as these modules are integrated into Backbone, the network's $F1$ and IoU are gradually improved, ultimately making EGAFNet achieve the best accuracy in this dataset. This shows that each module can function from different angles, thereby improving model accuracy overall.

2) Performance Analysis of Different Modules on a Dataset of Typical Urban Building Examples in China: The results of the ablation studies on the example dataset of buildings in typical Chinese cities are shown in Table IV. From Table IV, we can observe that after integrating the four modules (SAFM, EB, EGM, and MSM) into the Backbone network individually, the performance of the network shows different degrees of enhancement, which demonstrates that these modules can perform the corresponding functions according to the original design intention.

It is worth noting that when SAFM, EB, and MSM are added to the Backbone individually, the accuracy improvements are small, with $F1$ increasing by 0.17%, 0.28%, and 0.18%, respectively. However, when EGM is further added to EB, $F1$ increases to 85.25%, an increase of 0.95% compared to Backbone, which fully proves the positive impact of the introduction of boundary information on model accuracy. In addition, comparing rows 1, 2, 6, 7, and 8, we can find that when these modules are sequentially integrated into the backbone, the network's accuracy progressively improves, with the EGAFNet containing all modules naturally achieving the best accuracy. Therefore, these modules exhibit strong

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TABLE III
ABLATION STUDY RESULTS OF DIFFERENT MODULES ON THE WHU BUILDING DATASET

Backbone	SAFM	EB	EGM	MSM	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
√	×	×	×	×	97.75	92.30	94.17	93.23	87.31
√	√	×	×	×	97.87	92.85	94.31	93.58	87.94
√	×	√	×	×	97.88	93.89	93.20	93.54	87.87
√	×	√	√	×	98.00	93.87	93.95	93.91	88.51
√	×	×	×	√	97.92	**94.70**	92.50	93.59	87.95
√	√	√	×	×	97.89	93.14	94.08	93.61	87.98
√	√	√	√	×	98.05	93.39	94.85	94.11	88.88
√	√	√	√	√	**98.13**	93.82	**94.88**	**94.35**	**89.30**

TABLE IV
ABLATION STUDY RESULTS OF DIFFERENT MODULES ON A DATASET OF TYPICAL URBAN BUILDING EXAMPLES IN CHINA

Backbone	SAFM	EB	EGM	MSM	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
√	×	×	×	×	92.98	81.64	87.14	84.30	72.86
√	√	×	×	×	92.95	80.67	**88.63**	84.47	73.11
√	×	√	×	×	93.03	81.00	88.49	84.58	73.28
√	×	√	√	×	93.57	**84.59**	85.92	85.25	74.17
√	×	×	×	√	93.21	83.53	85.45	84.48	73.12
√	√	√	×	×	93.17	82.30	87.12	84.64	73.37
√	√	√	√	×	93.67	83.94	87.43	85.65	74.90
√	√	√	√	√	**93.84**	84.38	87.75	**86.03**	**75.49**

complementarity in practical applications. By combining these modules, EGAFNet not only overcomes the limitations of individual modules but also maximizes the advantages of each module, thereby achieving the greatest overall performance improvement. In summary, each module plays an indispensable role in enhancing the accuracy of EGAFNet.

3) Performance Analysis of Different Weighted Loss Combinations: In Section II-E, we elaborated on the loss function and set the weighting coefficients for each part of the loss. In this section, to explore the effect of different weighting coefficient combinations of the loss function on the performance of EGAFNet, we set up four different combinations of weighting coefficients and train EGAFNet while ensuring that the sum of the five weighting coefficients is 1. The experimental results are shown in Table V, where datasets A and B represent the WHU building dataset and a dataset of typical urban building examples in China, respectively. In each metric, the best results are shown in bold and the second-best results are underlined.

From the overall trend in Table V, the performance of the model shows an overall decreasing trend as λ_{seg} increases. It could be caused by L_{seg} corresponding to λ_{seg} dominating the model training, which makes the model focus too much on the segmentation task and neglect the tasks represented by other loss functions. In addition, the weights of L_{edge} also have some impact on the model performance. As the value of λ_{edge} decreases, the model performance also shows the same decreasing trend. It suggests that lower weight coefficients of L_{edge} may lead to a lack of sufficient boundary information for the building features in the decoding stage, which results in lower accuracy. In contrast, when all the weighting coefficients are equal, the model can consider each task more comprehensively and achieve the second-ranked accuracy on both datasets, showing a more favorable performance.

In summary, when designing the loss function weighting coefficients, the contribution of each task needs to be considered comprehensively to achieve the optimal performance. As shown in Table V, EGAFNet performs best with weighting coefficients of {0.1, 0.1, 0.1, 0.4, 0.3} on both datasets, so we have adopted this coefficient as the final parameter setting. However, the optimal coefficients may vary depending on the dataset and model, necessitating adjustments for different tasks and datasets in practical applications.

B. Model Complexity Analysis

Model complexity is an important measure of model performance, which reflects the extent of the model's demand for computational resources, including two main aspects: parameters and FLOPs. In Table VI, we list the model complexity of EGAFNet with other comparative methods. It is worth noting that to observe the balance more intuitively between complexity and accuracy, $F1$ of these methods on the WHU dataset was selected as the metric for model evaluation.

From the data in Table VI, it can be seen that the parameters of both methods, EfficientUNet+ and MSLNet, are in the top two positions, which is because both methods are constructed based on lightweight network. Although EGAFNet

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TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT WEIGHTED LOSS COMBINATIONS FOR THE TWO DATASETS

Dataset	$\{\lambda_1, \lambda_2, \lambda_3, \lambda_{seg}, \lambda_{edge}\}$	OA(%)	Precision(%)	Recall(%)	F1(%)	IoU(%)
A	{0.1,0.1,0.1,0.4,0.3}	**98.13**	**93.82**	94.88	**94.35**	**89.30**
	{0.1,0.1,0.1,0.5,0.2}	98.00	93.18	94.78	93.98	88.63
	{0.1,0.1,0.1,0.6,0.1}	97.99	93.55	94.30	93.92	88.54
	{0.2,0.2,0.2,0.2,0.2}	98.07	93.44	**94.90**	94.17	88.97
B	{0.1,0.1,0.1,0.4,0.3}	**93.84**	84.38	87.75	**86.03**	**75.49**
	{0.1,0.1,0.1,0.5,0.2}	93.71	**84.42**	86.94	85.66	74.92
	{0.1,0.1,0.1,0.6,0.1}	93.50	82.77	**88.31**	85.45	74.60
	{0.2,0.2,0.2,0.2,0.2}	93.77	83.82	88.17	85.94	75.35

TABLE VI
COMPARISON OF MODEL COMPLEXITY OF DIFFERENT METHODS

Method	Backbone	Parameters (M)	FLOPs (G)	F1 (%)
UANet	VGG-16	15.59	27.86	93.91
DSAT-Net	ConvNeXt	48.29	14.40	91.77
MSGCNet	ResNet-34	27.52	7.30	91.65
EfficientUNet+	EfficientNet	**5.76**	**2.68**	91.02
C^3Net	U-Net	73.67	71.18	93.28
MMBNet	Conv block	54.47	18.21	92.03
CFENet	ResNet-101	44.66	46.78	88.11
Res2-Unet	Res2Net	43.68	34.15	90.96
MSLNet	MobileNetV2	6.58	21.77	89.78
EGAFNet	EfficientNetV2	44.60	119.77	**94.35**

is also constructed based on the lightweight network, its parameters are ranked sixth, and the FLOPs rank last. This is because EGAFNet contains modules such as EB, EGM, and SAFM. These modules utilize a large number of convolution operations to extract image features, which increases the model complexity. Nonetheless, EGAFNet achieves better $F1$ and IoU compared to EfficientUNet+ and MSLNet. In addition, methods, such as DSAT-Net and MSGCNet, have less complexity due to the incorporation of transformer, but their accuracy is also inferior to EGAFNet. In general, the EGAFNet proposed in this article maintains good number of parameters while ensuring the accuracy of building extraction, but the FLOPs need to be reduced, so the focus of future work will be the design of a lightweight network to further enhance the performance and inference speed.

V. CONCLUSION

In this article, we proposed a novel EGAFNet for building extraction from high-resolution remote sensing images. The EGAFNet utilizes EfficientNetV2 as the encoder and constructs the MFEM module as the decoder to capture the deeper building features. To address the issue of missing building boundary information, we designed the EB to learn the boundary information and enhance the expression ability of the boundary features through the EGM module. In addition, considering the differences in the multiscale features, the SAFM module is embedded between the encoder and the decoder to capture the spatial context information at different scales, so as to enhance the ability of perceiving the buildings at different scales. Finally, the MSM module is introduced to predict the output of other decoder layer features and impose multilayer loss constraints, which further improves the accuracy of the model. The EGAFNet is experimentally validated on the WHU building dataset and a dataset of typical urban building examples in China. A series of comparative experiments and ablation studies demonstrate that EGAFNet achieves higher accuracy compared to other mainstream networks and exhibits better handling of boundary information, and the issue of insufficient integrity has been significantly improved. In future work, we will work on improving the efficiency of the model and exploring multimodal data fusion methods to provide more comprehensive and complementary information for building extraction tasks.

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Mingwang Yang received the bachelor's degree from Xuchang University, Xuchang, China, in 2021. He is currently pursuing with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, China.

His research interests include intelligent interpretation of remote sensing images and computer vision.

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Like Zhao received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2018.

She is currently an Associate Professor with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, Henan, China. Her research interests include photogrammetry, intelligent remote sensing, and image interpretation.



Linfeng Ye received the master's degree from Henan University of Technology, Zhengzhou, China, in 2024.

His research interests include intelligent interpretation of remote sensing images and computer vision.



Weishi Jia received the bachelor's degree from Liaoning Institute of Science and Engineering, Jinzhou, China, in 2022. He is currently pursuing with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, China.

His research interests include detection of wheat imperfections based on deep learning.



Huawei Jiang was born in Jiyuan, Henan, China, in 1970.

He is currently a Professor and a Doctoral Supervisor with the School of Information Science and Engineering, Henan University of Technology, Zhengzhou, China. His main research directions are optimization computing, intelligent information processing, and machine learning.



Zhen Yang received the Ph.D. degree in geographical information system from China University of Mining and Technology, Beijing, China, in 2019.

He is currently a Lecturer with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou, China. His research interests include digital image processing and remote sensing of resources and environment.

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